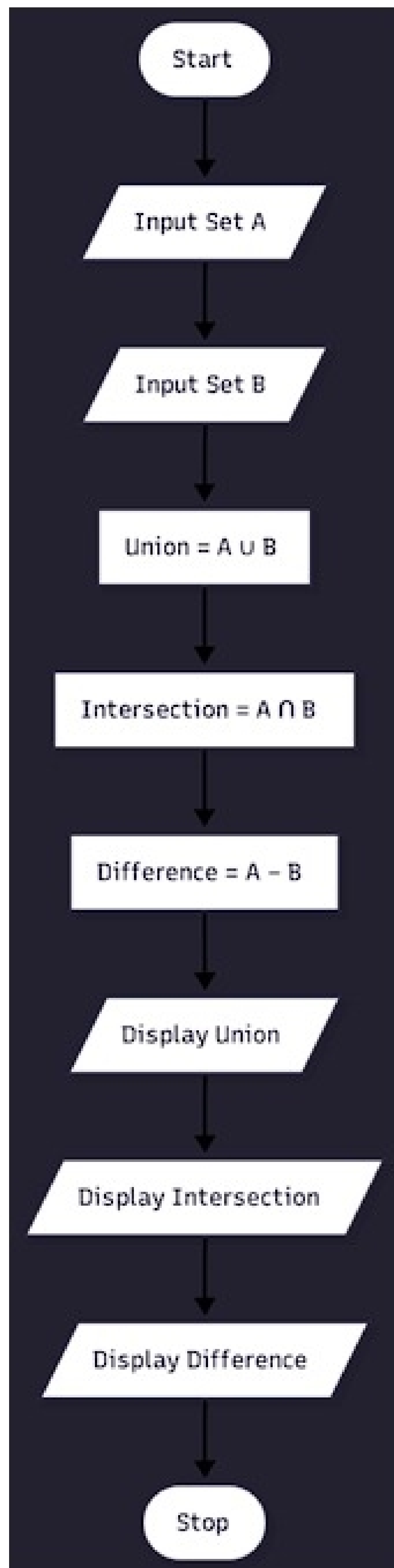


Algorithm: Set Operations (Union, Intersection, Difference)

1. **Start**
2. **Input Set A**
3. **Input Set B**
4. **Compute Union**
 - o Union = $A \cup B$
5. **Compute Intersection**
 - o Intersection = $A \cap B$
6. **Compute Difference**
 - o Difference = $A - B$
7. **Display Union**
8. **Display Intersection**
9. **Display Difference**
10. **Stop**



4.1.1. Set Operations

22:26



Write a Python program to perform union, intersection and difference operations on *Set A* and *Set B*.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for *Set A*.
- The second input prompts "Set B: " followed by space-separated list of integers for *Set B*.

Output Format:

- The first line prints "Union: " followed by the union of *Set A* and *Set B*.
- The second line prints "Intersection: " followed by the intersection of *Set A* and *Set B*.
- The third line prints "Difference: " followed by the difference of *Set A* and *Set B*.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases



setopera...

[Submit](#)

```
1 set_a = set(map(int, input("Set A: ").split()))
2 set_b = set(map(int, input("Set B: ").split()))
3
4 union_set = set_a | set_b
5 intersection_set = set_a & set_b
6 difference_set = set_a - set_b
7
8 print("Union:", union_set)
9 print("Intersection:", intersection_set)
10 print("Difference:", difference_set)
```

[Terminal](#)[Test cases](#)[< Prev](#)[Reset](#)[Submit](#)[Next >](#)